

Eigenvalues and inverse problems

The sharp off-diagonal threshold for spectral-subspace rotation

Difficulty: challenging **Importance:** interesting to the community**Status:** Open

Rating rationale: Challenging reflects an optimal off-diagonal perturbation bound for arbitrary separated spectral sets; community impact is a sharp guarantee for invariant-subspace stability.

Let A be a self-adjoint, possibly unbounded operator on a separable complex Hilbert space, with $\sigma(A) = \sigma \cup \Sigma$ for nonempty closed sets satisfying $d = \text{dist}(\sigma, \Sigma) > 0$. Let $P = E_A(\sigma)$. Suppose a bounded self-adjoint perturbation V is off-diagonal relative to this decomposition:

$$PVP = 0, \quad (I - P)V(I - P) = 0.$$

Writing $O_r(S) = \{x \in \mathbb{R} : \text{dist}(x, S) < r\}$ and $Q = E_{A+V}(O_{d/2}(\sigma))$, does

$$\|V\| < \frac{\sqrt{3}}{2}d \implies \|P - Q\| < 1$$

hold universally? Here E_B is the spectral projection measure of B , and norms are operator norms. No ordering of the two spectral sets is assumed.

The source retains the full operator setting, encompassing Hermitian block matrices in invariant-subspace computations. The threshold $\sqrt{3}/2$ already guarantees that the perturbed spectral components stay separated; the unresolved assertion controls their maximal subspace angle. Off-diagonal structure makes this a different threshold problem from the general-perturbation constant $1/2$.

References

A. Seelmann, *Notes on the subspace perturbation problem for off-diagonal perturbations*, Proceedings AMS 144 (2016), 3825–3832, §1, (1.1)–(1.4), and Theorem 2.5 ([primary manuscript; journal](#)). A. Seelmann, *Unifying the treatment of indefinite and semidefinite perturbations in the subspace perturbation problem*, Operators and Matrices 15 (2021), 1181–1188 ([primary paper](#)), treats a different extension of the perturbation hypotheses.

Status check — 2026-09-08

The 2016 paper explicitly conjectures $c_{\text{opt-off}} = \sqrt{3}/2$ and proves $c_{\text{opt-off}} > 0.6940725$. The 2021 extension does not attain this off-diagonal endpoint. Searches for “off-diagonal subspace perturbation”, “0.6940725”, “sqrt 3”, “optimal constant”, “conjecture”, and 2024–2026 found no resolution. This is a historical-source entry with a bounded later-literature check, not a certified claim of present openness.

Audit update — 2026-09-10

Rechecked [Seelmann’s account](#) and the [2021 follow-up](#). The proved sufficient constant above 0.694 remains below the conjectured threshold $\sqrt{3}/2$. Searches for subsequent off-diagonal optimal-threshold results found no general proof or counterexample. A smaller sufficient constant is progress but not a proved subfamily of the optimal-constant determination.